INCH-POUND

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MILITARY SPECIFICATION

TELESCOPE, ELBOW: 9360257

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for we by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a one and one half power (1.5X) fixed focus telescope known as Telescope, Elbow: 9360257 used with the U.S. Army mortar systems, M252. Throughout this specification this instrument is referenced as "telescope." This telescope has a radioactively illuminated reticle (see 3.2.1 and 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data swhich may be of use in improving this document should be addressed to: Commandr, U.S. Army ARDEC ATTN: SMCAR-GAGS Picatinny Arsenal New Jersey 07806-5000 by using the self-addressed Standardization Document Improvment Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-L-50778	Liquid Scintillation System for Radioactive Fire Control

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods and Enqineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless Unless otherwise specified, the Issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT ENGINEERING CENTER (ARDEC)

5549108	- Telescope, Collimating
8565556	- Pressure Tester Two Station
9360257	- Telescope, Elbow

PACKAGING DATA SHEET

PS9360257	- Packaging of Telescope, Elbow: 9360257

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DLAM 4145.8	- Radioactive Commodities in the DOD Supply system

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC. 20402.)

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DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory)
Part 20: Standards for Protection Against Radiation
Part 30: Rules of General Applicability to Licensing of By-product Material

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360257, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the telescope shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking
d. Workmanship.

3.2.1 Radioactive material. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, processing, handling, and shipping of radioactive material.

3.2.2 Eyeshield. The rubber eyeshield shall show no evidence of deterioration, cuts or tears and shall fit snugly on the telescope body.

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3.3 Environmental service conditions. Any telescope subjected to any or all of the requirements of 3.3.1, 3.3.2, 3.3.3, and 3.3.4 shall subsequently meet the requirements of 3.3.5 through 3.3.9 unless otherwise specified.

3.3.1 Vibration. The telescope shall be positioned with the line of sight through the eyepiece and objective tube in a horizontal plane and the telescope supported at the 0.624 inch diameter of the objective tube. The telescope shall then be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds.

3.3.2 Shock. The telescope shall be shocked six (6) times in each of two (2) positions. The applied shocks shall be inn a vertical, up direction. In the first position, the telescope shall be positioned with the line of sight through the eyepiece and objective tube in a horizontal plane. In the second position, the telescope shall be positioned with the line of sight through the eyepiece and objective tube in a vertical plane. In both positions the telescope shall be supported at the 0.624 inch diameter of the objective tube. The shock, for all twelve (12), shall be 300 g’s for one (1) mini-second and have a 1/2 sine wave pulse.

3.3.3 Temperature.

3.3.3.1 Operating. The telescope shall meet the requirements of 3.4 when stabilized thermally at any temperature between +60°F (+16°C) and +90°F (+32°C).

3.3.3.2 Nonopererating (storage). The telescope shall not suffer permanent damage when subjected to the constant temperature extremes of -50°F (-46°c) for a period of 24 hours and +160°F (+71°C) for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 when returned to room ambient.

3.3.4 Sealing and flushing. The telescope shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5 ± 0.10 psig for a period of two hours using dry nitrogen having a dew point at least as low as -25°F (-32°C). The telescope shall then be flushed with dry nitrogen until the interior atmosphere of the telescope has a dew point at least as low as -25°F (-32°c).

3.3.5 Condensation (low temperature). When viewed through the eyepiece and objective ends of the telescope, there shall be no evidence of condensation at room temperature immediately upon thermal stabilization, subsequent to being subjected to the -50°F (-46°C) and prior to the +160°F (+71°C) requirement of 3.3.3.2.

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3.3.6 Cleanliness. When viewed throught the eyepiece, the surface of the reticle shall show no more than six particles of dirt. The maximum width of each particle shall not exceed the apparent width of a reticle line. Teh separation between each dirt particle or foreign matter shall be at least 10 mils.

3.3.7 Defects. There shall be no evidence of moisture, grease, fingerprints, condensations, fractures, adhesive separations, or other forms of optical degradation on any glass component when the telescope is viewed through both the eyepiece and objective ends.

3.3.8 External tritium contamination. Tritium in any form shall not be present on the external surfaces of the telescope in excess of 1000 dpm/100 sq. cm.

3.3.9 Illumination check. Illuminated areas shall be given a visual check to determine if the light source is Illuminated.

3.4 Performance.

3.4.1 Collimation. The optical axis of the telescope shall coincide with the geometric axis within 1.0 mil when the telescope is rotated 3200 mils (90° to the left and 90° to the right from a . vertical position of the eyepiece) about the geometric axis as defined by the exterior mounting surface of the objective tube of the telescope.

3.4.2 Parallax. Parallax at the optical axis shall not exceed 0.5 mil when viewing a target between 40 meters and infinity.

3.4.3 Resolution. The resolution on the optical axis shall be 30 seconds of arc or less at one focus setting of an auxiliary observation telescope positioned at the eyepiece.

3.4.4 Parallelism of reticle and image. Teh verical line shall be parallel to the image of a plumb line within 1 degree of arc when each of the three indices (90° apart) are in position for coincidence with a vertical plane index through the geometric axis of the telescope's objective tube.

3.4.5 Eyepiece focus. The eyepiece focus of the reticle shall be set between minus .50 and minus 1.25 diopter.

3.4.6 Illiumination. The reticle lines shall be clearly distinguishable when observed, with and without a 50% neutral density filter, at light levels varied from zero to ten foot-candles.

3.5 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.6 Workmanship. The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which affect their protective qualities or apperance. Optical surfaces shall be clean and free of any scratches or blemishes which may affect their optical qualities and shall meet the requirements of 3.3.6 and 3.3.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract ofr purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchasw order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meetall requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspeciton, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classificaiton of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling. inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three telescopes for inspection In accordance with 4.3.2.

4.3.2 Inspections to be Performed. First article assemblies, components, and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II and Table III herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lof formation. The term "inspection lot" is defined as a homogeneous collection of units of Product from which a representative sample is drawn or which is inspected 100 percent to" determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspeciton. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, In one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the functional test paragraph. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table I below, using the inspection levels stated in -the functional test paragraph.

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TABLE I. Attributes sampling inspection

Lot Size	Inspection Levels			
	I	II	III	IV
2 to 8	*	*	*	*
9 to 15	*	*	*	13
16 to 25	*	*	*	13
26 to 50	*	*	32	13
51 to 90	*	*	32	13
91 to 150	*	125	32	13
151 to 280	*	125	32	32
281 to 500	*	125	32	32
501 to 1200	*	125	80	50

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, teh contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated Into the contractor's quality program or detailed inspection system, as applicable.

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4.4.2.1 Components, assemblies and subassemblies. All components, assemblies, and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed In the Technical Data Package (TDP). In the absence of QAPS, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (Individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection of item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests.
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	**None**	**N/A**
Major:		
101. Performance	**3.4**	**4.7.2**
Minor:	**None**	**N/A**

4.5 Special sampling. One telescope out of the first 50 produced and one out of each 100 produced thereafter shall be slected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

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TABLE III. Special sampling.

CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
Critical:	**None**	**N/A**
Major:		
102. Vibration	**3.3.1**	**4.7.1.1**
103. Shock	**3.3.2**	**4.7.1.2**
104. Temperature	**3.3.3**	**4.7.1.3**
105. Sealing	**3.3.4**	**4.7.1.4**
Minor:	**None**	**N/A**

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of the specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see.4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and

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permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Inspection equipment design requirements.

4.6.2.3.1 Special inspection equiment. Special testing equipment to test the performance of the telescope shall consist of a telescope holding device and a target projector collimator. The The test equipment shall conform to the following requirements:

a. Telescope holding device. This device shall be capable of securing and maintaining the exterior mounting surface of the telescope's horizontal objective tube within 20 seconds of arc. The device shall provide sufficient tension to the objective tube to permit it to be maintained in a "zero" rotational position while the telescope's mechanical axis lays in a vertical plane. This holding device shall also permit the telescope to be rotated through the excursion specified in 3.4.1 and 3.4.4 without slippage, or rotation under its own weight. This holding device shall also be equipped with three telescope position indices. One index shall indicate the "zero" rotational position of the telescope within plus of minus 8. minutes, the second index shall be positioned 90° ± 0°3' clockwise from the vertical plane, the third index shall be positioned 90° ± 0°3' counterclockwise from the vertical plane.

b. Target Prelector collimator. The target projector coil-ator shall have a clear aperture of at least 1.5 Inches. The The target reticle of the collimator shall contain a 1.0 mil diameter circle at the intersecting point of the horizontal and vertical reticle lines to determine conformance with collimation, as specified in 3.4.1. Tolerance marks shall also be incorporated on both sides of the vertical reticle line to determine conformance to parallelism of reticle and image, as specified in 3.4.4. Unless otherwise specified, the target collimator shall be set at infinity focus. The target collimator shall be positioned and aligned so that its optical axis is adjacent to, and parallel within 40 seconds to the geometric axis of the objective tube of the telescope under test. The vertical reticle line of the target collimator shall be adjusted plumb.

c. Sealing. Special testing equipment to test sealing of the telescope shall conform to the internal pressure test requirements of Drawing 8565556, Pressure Tester - Two Station. A special adapter shall be utilized when applying, or flushing, the nitrogen gas in the telescope.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers, and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. Conduct the vibration test with the telescope positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitude and frequency specified in 3.3.1. The applied motion shall be in a vertical plane for the time period specified. Upon completion of the vibration cycle the telescope shall be removed from the vibration adapter and subjected to the test as specified in 4.7.1.8, and then a visual examination for reticle illumination. It shall meet the requirements of 3.3.8 and 3.3.9, respectively. The telescope shall then be subjected to the tests as specified in 4.7.2.1 through 4.7.2.5 to determine overall conformance to the requirements of 3.4.

4.7.1.2 Shock. Conduct the shock test with the telescope positioned on an appropriate adapter affixed to a shock test machine that is capable of providing the shock as specified in 3.3.2. The adapter shall be capable of holding and positioning the telescope as specified in 3.3.2. The shock pulse shape, peak value and normal time duration shall be as specified In 3.3.2. subsequent ot the 12 shocks the telescope shall be removed from the shock test adapter and subjected to the test as specified in 4.7.1.8, and then a visual examination for reticle illumination. It shall meet the require-ments of 3.3.8 and 3.3.9, respectively. Teh telescope shall then be subjected to the tests in 4.7.2.1 through 4.7.2.5 to determine overall conformance to the requirements of 3.4.

4.7.1.3 Storage temperature. The telescope shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature to -50°F (-46°C). The telescope shall be maintained at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure the temperature of the chamber shall be raised, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The telescope shall then be removed from the test chamber and subjected to the tests specified in 4.2.1.8, and then a visual examination for reticle illumination. The telescope shall meet the requirements of 3.3.8 and 3.3.9, respectively. The telescope shall then be subjected to the test specified in 4.7.1.5 for evidence of moisture (condensation). The telescope shall again be placed in the test chamber and the temperature of the chamber raised, over a period of 4 hours ± 15 minutes, to +160°F (+71°C). The telescope shall remain at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure the temperature of

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the chamber shall be reduced, over a period of 4 hours ± 15 minutes, to room ambient temperature. The telescope shall then be removed from the test chamber and subjected to the test in 4.7.1.8, and then a visual examination for reticle Illumination. It shall meet the requirements of 3.3.8 and 3.3.9, respectively. The telescope shall then be subjected to the tests in 4.7.2.1 through 4.7.2.5 to determine overall conformance to the requirements of 3.4.

4.7.1.4 Sealing and flushing. Internal pressurizing of the telescope to test sealing shall be performed using dry nitrogen gas having a dew point at least as low as -25°F (-32°C) and special testing equipment conforming to 4.6.2.3.1. Teh telescope shall be pressurized to the requirements of 3.3.4 in accordance with the procedure for internal pressure testing of Drawing 8565556. Maintain the applied pressure for the period specified. With the pressure at 5 ± 0.1 psig, the telescope shall retain this pressure for 2 hours with no indication of leakage in excess of 0.2 psig as indicated on the pressure gage of the testing equipment. Upon satisfactory completion of this test, the body sealing screw shall be immediately assembled to the telescope as depicted on Drawing 9360257 to minimize intermingling of external atmosphere upon disconnection of the pressure tester adapter. The interior atmosphere of the telescope shall conform to the requirements of 3.3.4.

4.7.1.5 Condensation (low temperature). Subsequent to low temperature exposure and immediately upon stabilization at room ambient temperature, the telescope's optical system shall be examined as specified in 3.3.5 for evidence of condensation (moisture). This visual examination shall employ the technique of shadowing. The telescope shall conform to the requirements of 3.3.5.

4.7.1.6 Cleanliness. The optical surfaces of the telescope shall be inspected for cleanliness in accordance with MIL-O-13820. Inspection for dirt particles shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 foot-candles. Cleanliness of the telescope shall conform to the re-quirements of 3.3.6.

4.7.1.7 Defects. Conduct this test in sequence to the test in 4.7.1.6. Inspect for condensates,m grease, fingerprints, fractures, adhesive separations, and etc. This test shall be conducted with the unaided-eye utilizing the shadowing technique while viewing through the objective and eyepiece ends of the telescope. The optical system of the telescope shall conform to the requirements of 3.3.7.

4.7.1.8 External tritium contamination't. The contamination test shall be performed by wiping the item under test with filter paper moistened in distilled water. The damp filter paper shall be placed in a container with the proper percentage of scintillation liquid, which is determined by the type of scintillation test equipment used

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to determine the actual contamination. The scintillating system used to measure contamination must be calibrated and meet the minimum requirement of MIL-L-50778. Insert the sample into this system, count a minimum of 2000 disintegrations and convert the readout into microcuries (222 DPM 10^4=1 microcurie). The measured contamination shall be within the limits specified in 3.3.8. The liquid scintillating system must be calibrated, on six-month intervals, to a standard traceable to an N.B.S. standard.

4.7.2 Performance. Testing equipment utilized in the following tests shall conform to the requirements of 4.6.2.3.1.

4.7.2.1 Collimation. Position the telescope on the holding device with the eyepiece in a vertical position ("zero" rotational position). Observe through the eyepiece of the telescope. The intersecting point of the telescope reticle centerlines shall be in coincidence with respect to the intersecting point of the target reticle lines within the tolerance specified in 3.4.1 as determined by the 1.0 mil diameter circle of the target reticle. Repeat the above collimation test procedure with the telescope rotated 90° clockwise and 90° counterclockwise of the original vertical position. The intersecting point of the telescope reticle shall be within the 1.0 mil diameter circle of the target reticle at both positions to insure overall compliance with 3.4.1.

4.7.2.2 Parallax. Perform this test with the target collimator adjusted for 40 meters, then adjusted for infinity. Direct a line of sight through the telescope to the collimator target.Parallax is any apparent movement of the telescope reticle lines in relation to the target image when the observer’s head is moved from side to side or up and down. Patallax shall be measured on the optical axis of the telescope (center of the reticle pattern). Displacement between the telescope reticle and the target image shall not exceed the tolerance specified in 3.4.2.

4.7.2.3 Resolution. Position the telescope on a holding device capable of accepting the exterior mounting surface of the objective tube of the telescope. Perform this test utilizing an auxiliary telescope with a magnification of at least 3 power (3x) and a resolving power chart on a wall target. The test methods and procedures shall be in compliance with the "Resolution Test" specified in MIL-O-13830 using a resolving power chart that contains 4 line sets as shown in Figure 1 therein. The focus of the auxiliary telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count of the resolution target. Resolution of the telescope optical system shall meet the requirements specified in 3.4.3.

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4.7.2.4 Parallelism of reticle and image. perform this test in conjunction with the test in 4.7.2.1. Position the telescope with the eyepiece in the vertical position ("zero" rotational position). Adjust the telescope holding device in azimuth, if necessary, to position one extremity of the telescope's vertical reticle line coincident to the target vertical line image. Teh reticle line shall be parallel to the target vertical line image within the tolerance specified in 3.4.4, as measured at the other extremity of the telescope's reticle line with respect to the target image tolerance marks. Repeat the above test with the telescope shall meet the requirements of 3.4.4 in each of its 3 indexed positions.

4.7.2.5 Eyepiece focus. A calibrated dioptometer with a magnificaiton of at least 3 power (3x) and a diopter measuring range of zero to minus three diopter shall be utilized for measuring the eyepiece focus of the telescope reticle. Set the dioptometer objective diopter scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the eyepiece of the telescope. observe through the dioptometer, obtain best focus of the telescope reticle by adjustment of the dioptometer objective diopter scale. The telescope reticle shall be in best focus within the tolerance specified in 3.4.5, as measured on the dioptometer objective diopter scale.

4.7.2.6 Illumination. The reticle illumination test shall be performed under simulated conditions specified in 3.4.6. The luminosity test shall be performed in a dark room having a variably controlled light source. To perform this test, the observer must be dark adapted by being in a darkened room for a minimum of 15 minutes before viewing the reticle markings. Prior to performing the test, place a white matted surface approximately 2 feet from the telescope window. Perform the test by observing the reticle pattern and determine that this pattern is clearly discernible by means of the self-illuminating source in the dark room. Gradually raise the level of illumination to where the pattern becomes indiscernible. Center an aperture stop over the objective entrance aperture of the telescope and observe that the reticle pattern is clearly discernible. The aperture of the stop shall approximate one half (1/2) the interior diameter of the retainer for the objective lens. Raise the Raise the ambient level of illumination to 10 foot candles, remove the fixed stop and observe the reticle pattern. Repeat the above test with a 50% neutral density filter placed over the viewing end of the telescope. The reticle illumination of the telescope, with and without the 50% neutral density filter, shall meet the requirements of 3.4.6.

4.8 Workmanship. The telescope shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The telescope shall conform to the requirements of 3.6.

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS 9360257. The level of protection shall be as specified in the procurment document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Telescope, Elbow, 9360257, is used with the Mount Telescope, 9360169. These two items compose the Sightunit, 9360168, M64A1 which is used with the U.S. Army mortar systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).
c. Requirements for submission of first article sample.
d. Applicable stock number.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC), may also include drawings prepared by, and identified as U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.5 Subject term (key word) listing.

Fire Control System
Mortar
Sightunit

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of-the changes.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1240-AB38)

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